Wow an actual contents page

CONTENTS (The "Bore Zone")

It's a high altitude UAV (Insert cool name) → Bird related → Undead things.. Icarus is cool

To start... I want to study the updrafts influecing the growth of "Brads cloud"

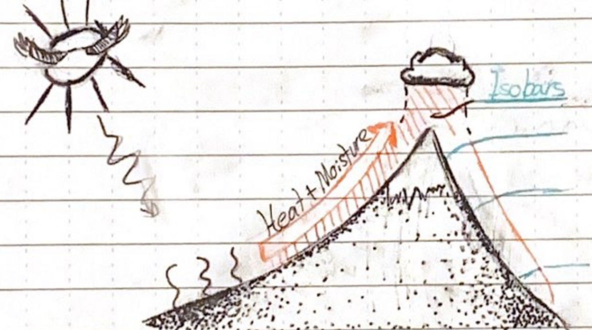
Fig1 My fever dream recreation of Brads cloud and Mt Taylor

The first challenge is getting an aircraft to 400ft then to 3,200 above launch site.

Second what sort of payload?

Lesson Time

Im dealing with a classic anabatic cloud. So hot air rises right? Well Brads cloud always appears early morning then grows.



This rising, hot, low pressure air is forced against the ridge. Once it reaches the top it meets the opposing slopes anabatic wind and boom. Cumulus cloud.

So payload?

- Camera on gimbal. Need to be able to look upslope
- GPS data to build an image of external conditions.
- Temperature gauge.
- A barometer. This allows the study of where the anabatic zone is
- 3 Axis wind direction + speed.
- Humdity is huge
- Ground scaning radar ??
- IMU
- Thermal camera

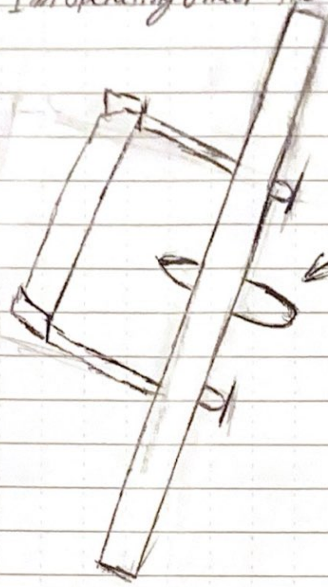
And total flight time 90 min

2 So... its like a medium altitude drone ...

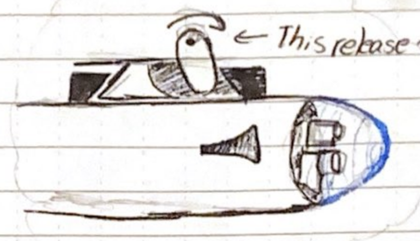
Wing Design

I got a couple ideas for the general layout. Firstly twin boom

I'm operating under the sensor pod ejection system.

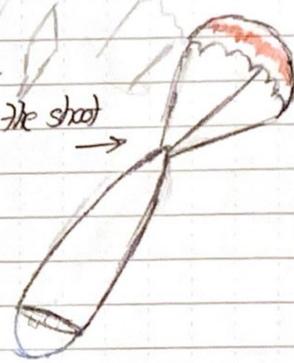


So in an emergency

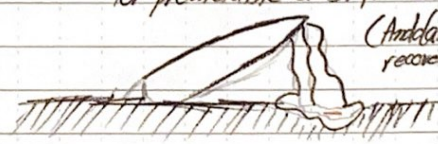


← This release mech

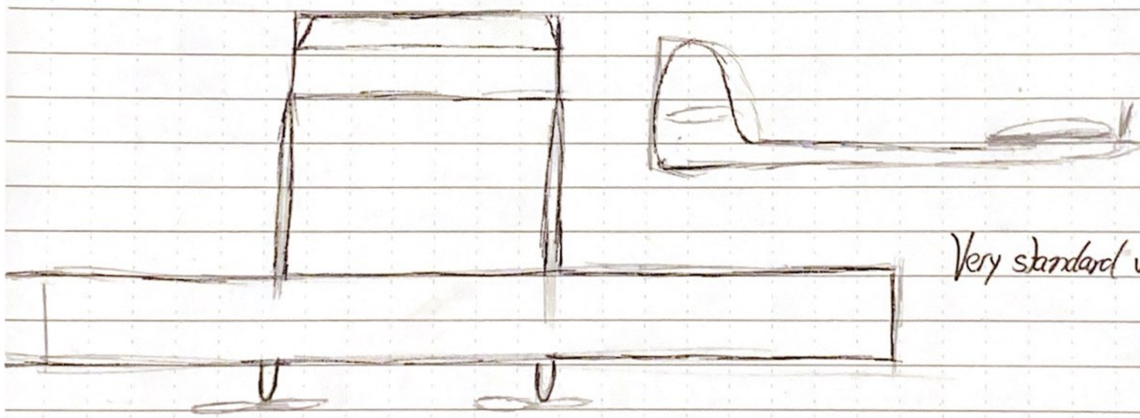
Frees the sensor pod and pulls the shoot →



For predictable crash prediction (And data recovery)

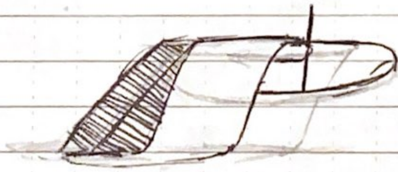


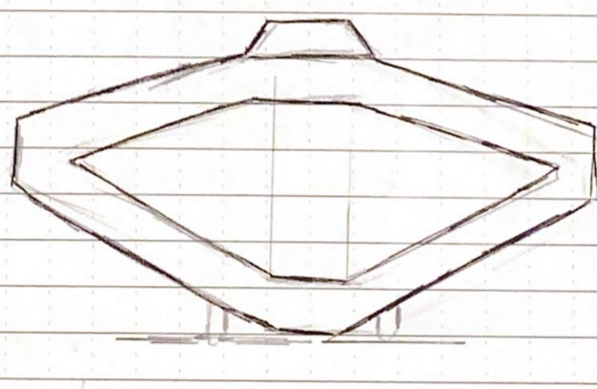
#1 So the standard wing config...



Very standard w/ foldable props.

2# My favorite the closed loop wing





The next big thing to settle is which is more efficient. From the paper "Investigation of flight conditions where Box-wing Outperforms Mono-Wing Configurations for small UAVs" by J. Aloor States "box wings outperformed mono-wings at elevated velocities and with increased lift demands".

Wing ridgitity

The main option or outcome that makes this box wing config appealing is the damping that the stucture provides. Flying in turbulence is the name of the game. So... the next step is to validate these results from Aloor

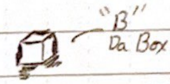
Aircraft design: The basics

Pressure:

As you exist in the world air exerts pressure on your body. Your body exerts a force back as well. The definition of pressure here is

$$p = \lim \left(\frac{dF}{dA}\right) \; dA \to 0 \; ; \; dA = \text{area}, \; dF = \text{Force on one side of } dA$$

So some object with a VERY small area exerts a force described as Force per unit area



Density:

We can define density in a very similar manner.

$$\rho = \lim \left(\frac{dm}{dV}\right) \; dV \to 0 \; ; \; dm \text{ is the mass of fluid inside } dV$$

dV is the volume of fluid around B

Flow velocity:

This is the vector path a particle "B" flows along stream lines.

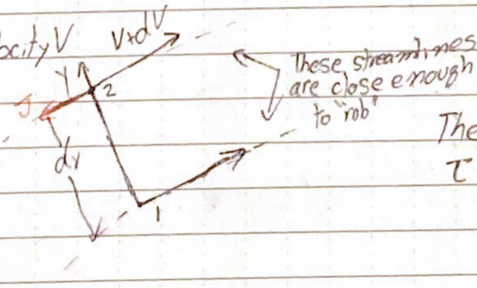
Stream lines:

Speaking of streamlines, these are lines of steady flow. These assist with visuallizing the motion of gas.

Friction: (Shear stress)

Air like any other fluid has to interact with itself. Lets setup 2 different streamlines extremely close together

Point 1 is traveling with velocity V
Point 2 is slightly faster at $V + dV$



The shear stress at point 1 is

$$\tau = \lim \left(\frac{dF_f}{dA}\right) ; \; dA \to 0$$

Scale: 1 square = _______

Rite in the Rain

Shear acts tangentially along the streamline. In terms of ~~veloci~~ viscosity, μ, shear can be described as

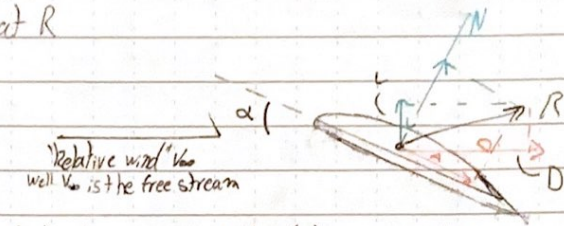
$$\tau = \mu \cdot \frac{dV}{dy} \text{ - Velocity gradient}$$

Aerodynamic Forces + Moments

There are two aerodynamic forces acting on a body:

1. Pressure distribution over the body (p)
2. Shear stress distribution over the bodies surface (τ)

Pressure is normal to the body and shear is tangental. Integrating the forces over the body produce the force vectors R and M. R is the aerodynamic force and M is the moment. Taking a look at R

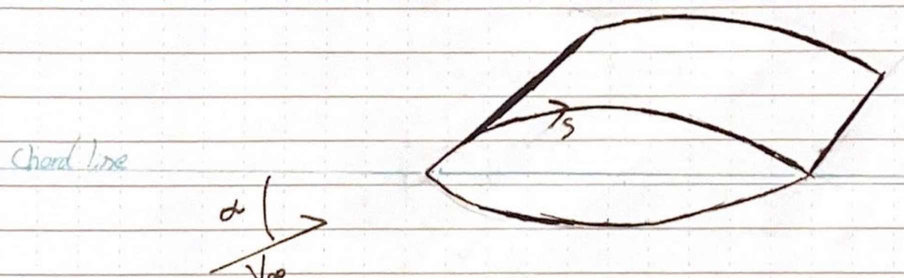


The angle of attack is α and it determines lift (L) and drag ratios (D))

*Angle of attack is clockwise to freestream angle

$$L = N\cos\alpha - A\sin\alpha$$
$$D = N\sin\alpha + A\cos\alpha$$

N is the normal force and A is the axial. You remember what acts normally and tangential. Pressure is normal and shear is axial.



Were going to examine the wing surface by breaking "s" into bits and adding their vector contributions. So the infinitesemal contributions of Normal and Axial are respectivly

$$dN'_{upper} = -p_{upper}\, ds_{upper} \cos\theta - \tau_{upper}\, ds_u \sin\theta$$

$$dA'_u = -p_u\, ds_u \sin\theta + \tau_u\, ds_u \cos\theta$$

And the lower,

$$dN'_l = p_l\, ds_l \cos\theta - \tau_l\, ds_l \sin\theta$$

$$dA'_l = p_l\, ds_l \sin\theta + \tau_l\, ds_l \cos\theta$$

Moments

Aircraft stability is contributed heavily by the wings moment. This is the final piece to calculate $p(s)$ and $\tau(s)$ to determine how the wing performs aerodynamically.

Instead of deriving these contributions are given

$$dM_u' = (p_u \cos\theta + \tau_u \sin\theta) x\, ds_u + (-p_u \sin\theta + \tau_u \cos\theta) y\, ds_u$$

$$dM_l' = (-p_l \cos\theta + \tau_l \sin\theta) x\, ds_l + (p_l \sin\theta + \tau_l \cos\theta) y\, ds_l$$

Performance coeff.

The dynamic pressure describes the atmospheric pressure far ahead of the body in the freestream

$$q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2$$

This takes us to the list

$C_L \equiv \dfrac{L}{q_\infty S}$ Lift coef.

$C_D \equiv \dfrac{D}{q_\infty S}$ Drag

S is the cross section platform area

c is the Chord length Also l

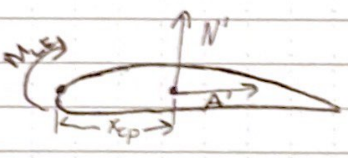
$C_N \equiv \dfrac{N}{q_\infty S}$ Normal force

$C_A \equiv \dfrac{A}{q_\infty S}$ Axial force

$C_M \equiv \dfrac{M}{q_\infty S l}$ Moment coef.

Center of Pressure

Alternitively the center of pressure is where the aerodynamic moment is zero.



The length x_{cp} is found with

$$x_{cp} = -\frac{M_{LE}'}{N'}$$

At low angles of attack the equation $L = N\cos\alpha - A\sin\alpha$ becomes $L \approx N$

So Box or Standard?

Aloore used a fairly limited number of test cases. Critically in Box wings the rear stabalizer is integrated and in standard configs the rear (wow lost any train of thought). The rear is not integrated in standard so Aloors A+B test is only valid for flying wings.

1st Up I need to reproduce Aloores MGV result.

Scale: 1 square = ________

		Span	Area	AR	Re	V
M6V	Mean Aerodynamic Cord 70mm	400mm	$0.028 m^2$	6	1.5×10^5	22.1 m/s

So for my version of M6V

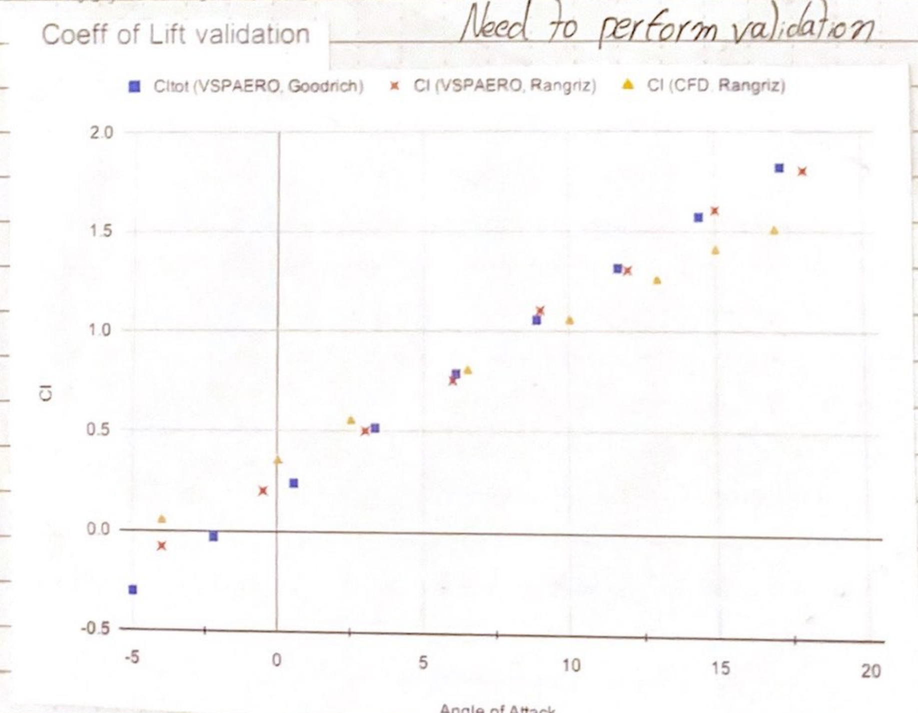
AR = 6.0
Span = 0.40988 m
Area = 0.02800
Taper = 0.750
Ave C =

Second try

AR = 6.0 | I think the Re might be messed up or mach #
Span = .420 |
Area = 0.02440 | Maybe... I'm getting extremely low values for my CD
Taper = .750 | Airfoil is wrong
Ave C = 0.070 |
Root C = 0.080 | Correct airfoil $\alpha = 5°$ CLtot ≈ 0.00043
Tip C = 0.060 | $\alpha = 10°$ CLtot ≈ 0.00068
Sec 22.71851 |
Mach: 0.064548 Charging Re to 1.5×10^7 CLtot ≈ 0.00068 $\alpha = 10°$
α = 5° Reseting Re to 1.5×10^5, CLtot ≈ 0.00068 $\alpha = 10°$

Refference area must be set to target!
CL tot ≈ 0.73278 @ 1.5×10^5 $\alpha = 5°$
CD tot = 0.02364
L/D = 30.99
Adjusting to mach 0.065
CL tot = 0.73171

Need to perform validation



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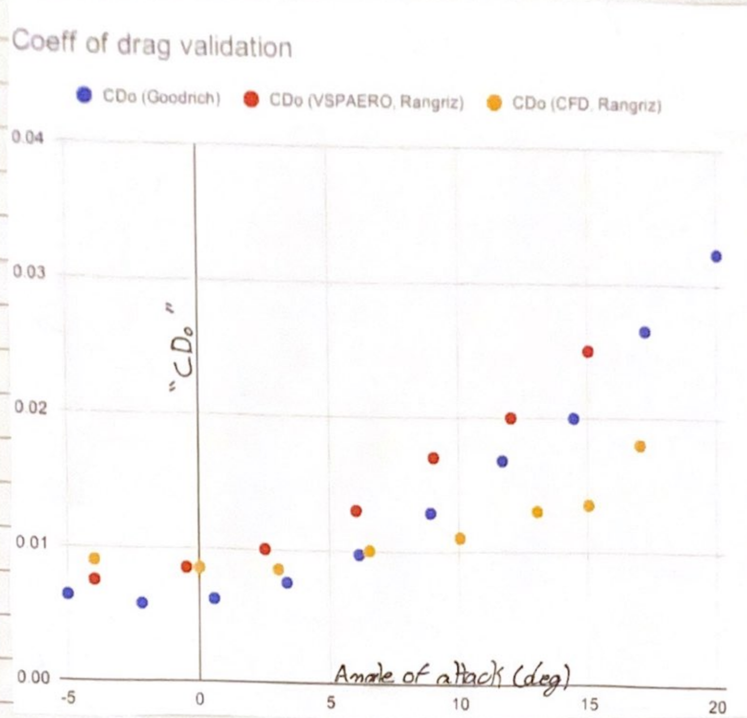


So my results strongly agree with Rangriz. But not Aloore

A Bit on OPENVSP

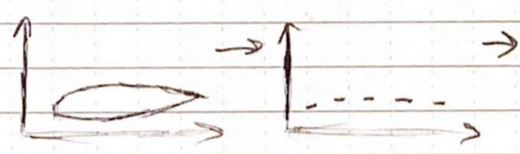
VLM - Vortex Lattice method

So wheres the vortex, hows it on a lattice and where is the method

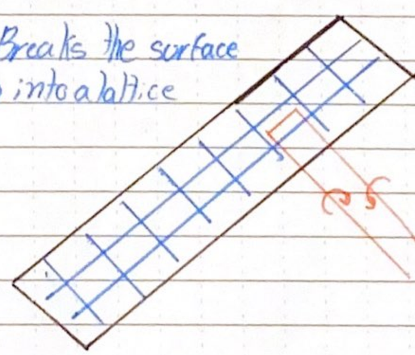
From "Fundamentals of aerodynamics": Consider a flow where all stream lines are concentric circles about a given point, ..., with the velocity in each circular streamline constant, but varying from one to another inversely with the distance from the common center.

And with that I realize there is much more to the topic. In short..

The VLM



Takes the air-foil geometry → Finds its equivalen thin form → Breaks the surface up into a lattice



Then computes these horse shoe vortex over the lattice